Canonical Multichart Structural Decomposition in the Competitive Regime

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Abstract

In this work, I present V4 as the canonical multichart/multicomponent extension of the structural theory. Mirroring the architecture of the V3 manuscript, I define the structural variables, describe the mixture decomposition, introduce the multicomponent variational functional, state local identifiability and robustness propositions, formalize the ontological transition criterion $\chi_{3\to 4}$, and reorganize the computational evidence with concrete tables and references. The manuscript is scoped to the present state of the project, but its central point is already precise: the external gauge is no longer left at the level of solver heuristics and is instead specified through an admissible external class, a finite canonical fusion operator, a reduced competitive class, a lexicographic canonical selector, and a formal fallback rule to V3.

1 Introduction

V4 enters once the affine core (V2) and the nonlinear single-chart regime (V3) have stabilized the canonical gauge for densities with simple global geometry. Benchmarks such as `cisao_em_quatro`, `sela_torcida`, and `multimodal_leve` expose densities whose residual fragmentation, Jacobian pressure, or multicart gains persist even after selecting the canonical V3 chart. In these regimes a single chart ceases to be the correct ontology: either the Jacobian degenerates, the residual remains genuinely fragmented, or a systematic multichart gain indicates that one branch is no longer enough.

V4 therefore upgrades the language from "density = one chart + residual" to "density = finite mixture of structural components." The purpose of the present document is to record the current coherent state of the theory and, in particular, to close the external gauge more sharply than in the exploratory stage: V4 already exists as a principled ontological extension beyond V3, has a working solver with a canonical choice of K, and now admits a deterministic external gauge built from admissibility, fusion, competitiveness, and lexicographic selection.

2 Structural Variables and Decomposition

2.1 Structural Variables

I model a density $\rho(x) \geq 0$ through a finite mixture

$$\rho = M \sum_{a=1}^{K} \alpha_a (\Phi_a)_{\#} \eta_a, \qquad \sum_{a=1}^{K} \alpha_a = 1, \qquad \alpha_a \geq 0,$$

where M is the total mass. Each component $(\alpha_a, \Phi_a, \eta_a)$ carries structural variables mirroring the canonical pieces of V3:

- $\alpha_a M$ is the structural mass of component a;

- $\Phi_a(\xi) = q_a + A_a\xi + N_a(\xi)$ is the local chart, with center q_a, SPD anchor A_a, and nonlinear correction N_a;
- η_a is the normalized residual profile satisfying

$$\int \eta_a = 1, \qquad \int \xi\, \eta_a(\xi)\, d\xi = 0, \qquad \int \xi \otimes \xi\, \eta_a(\xi)\, d\xi = \Sigma_a.$$

Each component thus inherits the internal language of V3, while V4 adds an external gauge deciding how many such components are ontologically necessary.

2.2 Structural Decomposition

The mixture can be read either as a finite atlas of local charts or as a structural mixture of branches. The present manuscript adopts the mixture viewpoint as the primary one, because it is the natural language of the current solver and of the current validation suites. The atlas interpretation remains the geometric background: each component describes one coherent branch of the observed density, and the external gauge decides when two branches are genuinely distinct and when they are merely redundant duplications of the same structural object.

3 Variational Multicomponent Functional

3.1 Primal Functional

$$\mathcal{J}_{V4}(\Theta) = \sum_{a=1}^{K} \alpha_a \mathcal{J}_{V3}^{(a)} + \mu\mathcal{E}_{\text{sep}} + \nu\mathcal{E}_{\text{merge}} + \lambda\mathcal{E}_{\text{comp}} + \kappa K.$$

Each component contributes its local V3 structural action $\mathcal{J}_{V3}^{(a)}$, while the new penalties regulate the multicart architecture:

- $\mathcal{E}_{\text{sep}}$ enforces structural separation between components;
- $\mathcal{E}_{\text{merge}}$ penalizes redundant or nearly duplicated branches;
- $\mathcal{E}_{\text{comp}}$ regularizes higher-order complexity;
- κK penalizes unnecessary growth in the number of components.

The role of this functional is not merely to reduce reconstruction error. It also organizes the external gauge by balancing fidelity and parsimony, so that V4 remains a theory of multichart necessity rather than a free mixture fit.

3.2 Canonical Choice of K

The number of components K is part of the external gauge. The canonical rule is:

(1) generate candidate decompositions for $K \in \{2, 3, 4\}$;

(2) compute the best reconstruction error for each K;

(3) keep only the values of K whose error remains competitive within the calibrated tolerance τ_K;

(4) choose the smallest competitive K;

(5) apply canonical fusion and non-redundancy before declaring the final ontology.

In the current implementation, τ_K is calibrated scenario by scenario from the dedicated validation suite. This already makes the gauge quantitative, and the external layer is closed by packaging the rule into an admissible class, a canonical fusion operator, and a final lexicographic selector.

3.3 External Admissibility, Fusion, and Reduction

Fix thresholds

$$(\tau_K, \varepsilon_{\text{cent}}, \varepsilon_{\text{pred}}, m_{\min}).$$

I call a multichart decomposition

$$\Theta = (M, K, \{(\alpha_a, \eta_a, \Phi_a)\}_{a=1}^{K})$$

externally admissible if:

(i) the components are ordered by mass, $\alpha_1 \geq \cdots \geq \alpha_K$;

(ii) the masses satisfy $\alpha_a \geq 0$ and $\sum_a \alpha_a = 1$;

(iii) each local component belongs to the V3 admissible class;

(iv) for each pair $a \neq b$, either the center distance exceeds $\varepsilon_{\text{cent}}$ or the prediction distance exceeds $\varepsilon_{\text{pred}}$;

(v) each component satisfies the minimum structural mass bound $\alpha_a \geq m_{\min}$.

I then define the canonical fusion operator $\mathfrak{F}_{\text{can}}$ as the deterministic iteration that merges the lexicographically nearest admissible pair whenever both distances fall below threshold, or whenever one component has mass below $m_{\min}$, then reorders by mass and repeats. Since each merge strictly decreases the number of components, $\mathfrak{F}_{\text{can}}$ terminates after finitely many steps. A decomposition is called *reduced* when

$$\mathfrak{F}_{\text{can}}(\Theta) = \Theta.$$

Proposition 1 (Canonical External Gauge)**.** *Fix the internal V3 gauge, the thresholds*

$$(\tau_K, \varepsilon_{\text{cent}}, \varepsilon_{\text{pred}}, m_{\min}),$$

and the candidate set $\mathcal{K}$. For each $K \in \mathcal{K}$, let $\Theta_K^\star$ be an optimal reduced decomposition. Define

$$E_* = \min_{K \in \mathcal{K}} E_{\text{rec}}(\Theta_K^\star),$$

and let the competitive class be

$$\mathcal{C}_{\tau_K} = \{\Theta_K^\star : E_{\text{rec}}(\Theta_K^\star) \leq (1 + \tau_K) E_*\}.$$

Then the external gauge is the lexicographic minimizer

$$\mathfrak{G}_{\text{ext}}(\rho) = \operatorname*{LexMin}_{\Theta \in \mathcal{C}_{\tau_K}} \Big(K(\Theta),\, \mathcal{J}_{V4}(\Theta),\, (\alpha_1, \ldots, \alpha_K),\, \mathbf{d}(\Theta) \Big),$$

where $\mathbf{d}(\Theta)$ is the ordered vector of residual structural distances between surviving components. In particular, once the thresholds and internal gauge are fixed, the external gauge is canonically determined.

Sketch. The fusion operator terminates in finitely many steps because each step lowers K. For each candidate K, fusion and reordering produce a reduced representative. The competitive class retains only decompositions that are reconstruction-competitive. The lexicographic selector then removes the remaining ambiguity by minimizing, in order, complexity, structural objective, mass profile, and residual distances. Therefore the external layer is no longer just a solver preference; it is a deterministic canonical operator on reduced decompositions. □

4 Propagation and PDE Interpretation

The next theoretical step is to show that, once the multichart gauge is established at an initial time, it propagates coherently under the same transport-reaction dynamics used in V3. What is established here is the local propagation mechanism naturally induced by the V4 structure; a fully global PDE theory remains outside the scope of this manuscript.

Proposition 2 (Propagation Template for the Multichart Gauge)**.** *Let*

$$\rho_0 = \sum_{a=1}^{K} \alpha_a (\Phi_a)_{\#} \eta_a$$

be an admissible multichart decomposition satisfying the external gauge constraints. Assume the evolution is driven by admissible velocity fields whose gradients remain uniformly bounded on a short time interval. Then, for sufficiently small time, there exists a propagated decomposition with the same number of components, the same qualitative gauge structure, and controlled variation of the charts, masses, and separation penalties.

Sketch. The argument follows the same local logic as in V3. Bounded gradients imply local Lipschitz control on each propagated chart, so the SPD anchors remain nondegenerate for short time. The distances between centers and the component predictions vary at most linearly in time, which keeps the separation and merge penalties within the admissible range. Therefore the multichart gauge survives locally as long as the thresholds are not crossed. □

Theorem 1 (Short-Time Preservation Under Transport-Reaction Dynamics)**.** *Assume $\rho(x,t)$ solves an admissible transport-reaction equation with bounded forcing and bounded velocity gradient, and assume $\rho(\cdot,0)$ admits an admissible V4 decomposition satisfying the current gauge thresholds. Then there exists $T' > 0$ such that for all $t \in [0,T']$ the solution still admits a V4 decomposition with controlled charts, controlled residual normalization, and preserved admissibility thresholds.*

Sketch. Mass variation is controlled by the forcing term, while the chart deformation remains controlled by the velocity gradient. The Jacobian stays positive for short time because the SPD anchors start away from degeneracy and vary continuously. The residual normalization and merge penalties receive only small perturbative corrections on a short interval. Combining these bounds yields the claimed persistence of admissibility. □

5 Local Identifiability and Robustness

The V4 analogue of the V3 local theory is formulated here at the local level relevant to canonical gauge selection.

Proposition 3 (Local Identifiability)**.** *Under the canonical gauge, fixed K_{can}, and enforced separation thresholds, the multicomponent decomposition is locally identifiable up to relabeling: sufficiently small perturbations of the data induce unique nearby parameters $(\alpha_a, \Phi_a, \eta_a)$ within the same gauge class.*

Sketch. One applies the V3 local identifiability argument component-wise and uses the separation penalty to prevent collisions between branches. Once branch collisions are excluded, the reduced Hessian remains locally nondegenerate on each component, and only the usual label symmetry remains. □

Proposition 4 (Local Robustness)**.** *Under fixed thresholds (τ_K, κ) and canonical fusion, perturbations of the data or changes of initialization yield decompositions that remain in controlled neighborhoods of the original parameters, with bounded variation in K and without uncontrolled explosion of residual fragmentation.*

Sketch. The thresholds act as barriers. Nearly duplicated branches are absorbed by fusion, and the complexity penalty prevents sudden growth of K unless reconstruction competitiveness justifies it. This reproduces, at the multichart level, the same local robustness philosophy already used in V3. □

6 Ontological Limit and Transition Criterion

6.1 Criterion

The V3→V4 transition is encoded in the object

$$\mathsf{Criteria}_{3\to 4} = \left\{ E_{V3}, E_{V4}, G_{\text{gain}}, J_{\text{jac}}, F_{\text{res}}, \tau_{\text{strong}}, \kappa_{\text{frag}}, \tau_{\text{jac}}, \kappa_{\text{jac}}, \kappa_{\text{frag}}^{\text{acpl}} \right\},$$

where

$$G_{\text{gain}} = \frac{E_{V3} - E_{V4}}{\max(E_{V3}, \varepsilon)}$$

measures multichart gain, J_{jac} measures Jacobian pressure in the best V3 chart, and F_{res} measures residual fragmentation.

The boolean $\chi_{3\to 4}$ is triggered when strong gain plus fragmentation is present, or when a moderate gain is coupled to both Jacobian pressure and residual fragmentation. This criterion is external: it diagnoses ontological failure of the single-chart regime before the full V4 search is accepted.

6.2 Fallback Rule

If $\chi_{3\to 4}$ fires, `solve_v4_theoretical` competes candidate decompositions for $K \in \{2, 3, 4\}$, applies canonical fusion and non-redundancy, and returns the smallest competitive multichart ontology. If fusion collapses the solution to $K_{\text{can}} = 1$, the correct reading is a formal return to V3. Thus V4 is not an always-on enlargement; it is the regime entered precisely when single-chart ontology becomes insufficient.

Proposition 5 (Formal Fallback to V3)**.** *Let $\Theta_{\text{can}} = \mathfrak{G}_{\text{ext}}(\rho)$ be the canonical V4 decomposition. If*

$$K(\Theta_{\text{can}}) = 1,$$

then the correct ontological label for ρ is the single-chart regime, and the structural description must be read as belonging to V3 rather than to V4.

Sketch. By construction, $\mathfrak{G}_{\text{ext}}$ already incorporates competitiveness, fusion, reduction, and lexicographic minimality. Therefore, if the canonical representative has only one surviving component, no competitive nonredundant multichart ontology remains. The multichart language has eliminated itself, so the correct regime is V3. □

Theorem 2 (Structural Cost Threshold)**.** *Let ρ admit admissible V3 and V4 descriptions. Define*

$$\mathcal{C}(\rho) = (J_{\text{jac}}(\rho), F_{\text{res}}(\rho), G_{\text{gain}}(\rho), \Delta_{\text{merge}}(\rho)) .$$

There exist thresholds

$$(\kappa_{\text{jac}}, \kappa_{\text{frag}}, \tau_{\text{strong}}, \delta_{\text{merge}})$$

such that:

(i) if at least one relevant structural cost exceeds its threshold, the single-chart regime becomes structurally insufficient and $\chi_{3\to 4} = 1$;

(ii) if all costs stay below threshold and canonical fusion collapses the multichart competitor to one branch, then $\chi_{3\to 4} = 0$ and V3 remains the correct ontology.

Sketch. The thresholds are read off from admissibility: Jacobian collapse is limited by κ_{jac}, removable fragmentation by κ_{frag}, competitive gain by τ_{strong}, and branch collision by δ_{merge}. Once one of these barriers is crossed, the single-chart representation can no longer keep all structural costs under control. Conversely, if the barriers are respected and fusion retracts the multichart candidate to one branch, no ontological reason remains to leave V3. The thresholds are calibrated from the validation suites, which makes the transition criterion quantitative inside the present theory. □

7 Computational Evidence

The following tables summarize the validation suites after the latest runs in the repository.

7.1 Summary Table

Suite	Reference	Outcome
Solver validation	`V4/validate_v4_solver.py`	60 cases, proxy error 1.302, solver error 5.514
Transition vs V3	`V4/transition_vs_v3_validation.py`	240 cases, mean errors 2.493/1.649/0.603 for
Tau calibration	`V4/calibrate_tau_k.py`	calibrated τ_K, including 0.2247 for `dobrament`
Gauge stability	`V4/gauge_stability_validation.py`	seed: $\text{std}_K = 0.029$, coassignment 96.85%; per
Guided restarts	`V4/saddle_guided_restart_validation.py`	lower objectives and improved convergence in

7.2 Error and Gain Table

Comparison	Baseline error	V4 error	Relative gain
Proxy vs solver	1.302	0.551	36.98%
V3 quadratica vs V4	2.493	0.603	75.81%
V3 cubica vs V4	1.649	0.603	63.43%

7.3 Tau Calibration Map

Scenario key	Calibrated τ_K
`dobramento_senoidal`	0.2247
`cisao_em_quatro`	0.0000
`anel_filamentado`	0.3755
`sela_torcida`	0.0000
`multimodal_leve`	0.0931

7.4 Stability Table

Mode	std_K	Coassignment	Quasi $\chi_{3\to4}$	Convergence
Seed	0.029	96.85%	85.83%	14.17%
Perturb	0.103	88.38%	83.33%	16.67%

7.5 Narrative

Taken together, these suites support the claim that the transition criterion and the external gauge remain meaningful beyond the proxy stage. The solver validation shows that V4 already functions as its own regime rather than as a mere inherited heuristic. The transition suite shows that V4 dominates the stable V3 quadratica layer and also the stronger V3 cubic layer in the regimes where fragmentation is structurally relevant. The tau calibration suite explains why extra branches are retained only when they remain competitively necessary. The gauge stability suite shows that the same reduced ontology typically reappears under seed and perturbation changes, which is the stability notion required by the theory.

7.6 Current Limitations

The same evidence identifies the present frontier of V4. First, strict convergence is still modest, so the solver remains dependent on restarts and initialization heuristics. Second, threshold calibration remains scenario-sensitive, so an abstract theory of universal threshold choice would still be valuable. Third, `anel_filamentado` continues to be the clearest counterexample to a fully stabilized gauge: it is improved relative to V3 in transition tests, but still exposes instability and model-selection fragility in the dedicated solver validation. These limits do not weaken the definition of the external gauge; they mark where its quantitative control remains hardest.

8 Conclusion

V4 now has a documented ontology, a workable internal/external gauge architecture, a variational organizing functional, local identifiability and robustness principles, an operational transition criterion, and computational pipelines aligned with the previous manuscripts. More importantly, the external gauge is no longer described only operationally: it is now specified by an admissible class, a finite canonical fusion operator, a reduced competitive class, and a lexicographic canonical selector, together with a formal fallback rule to V3. The present document therefore mirrors V3 while making precise the ontological switch that justifies a multichart description and the mathematical mechanism by which the multichart ontology is selected.

References

- S. Chizat *et al.*, references on unbalanced optimal transport (Wasserstein-Fisher-Rao, Hellinger-Kantorovich) for comparative discussion.
- Internal structural references: `main.tex`, `V3/v3_manuscript.tex`, `V4/knowledge_v4.md`, and `V4/v4_foundation.md`.
- Computational scripts supporting the tables and ontological claims:
  - `V4/validate_v4_solver.py`
  - `V4/transition_vs_v3_validation.py`
  - `V4/calibrate_tau_k.py`
  - `V4/gauge_stability_validation.py`
  - `V4/saddle_guided_restart_validation.py`

Declarations

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Code availability The scripts cited herein — `V4/validate_v4_solver.py`, `V4/transition_vs_v3_validation`, `V4/calibrate_tau_k.py`, `V4/gauge_stability_validation.py`, and `V4/saddle_guided_restart_validation.py` — are provided alongside the manuscript and are made available under the GNU AGPLv3 license in the project repository.